



BHABHA UNIVERSITY ,BHOPAL

Scheme of Examination

Third Semester- Diploma in Engineering (Electrical Engineering) Session 2019-20

SPECIALIZATION – Electrical Engineering

S.No.	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem. Exam.	Tests (Two)	Assignments /Quiz	End Sem. Practical /Viva	Practical Record/ assignment/Quiz /Presentation	
1	EE-3001	ELECTRICAL MACHINES – I	4	1	2	6	70	20	10	30	20	150
2	EE-3002	ELECT. & ELECTRONICS MEASUREMENT AND MEASURING INSTRUMENTS	4	1	2	6	70	20	10	30	20	150
3	EE-3003	BASIC ELECTRICAL ENGG. & MATERIALS	4	1	2	6	70	20	10	30	20	150
4	EE-3004	BASIC ELECTRONICS	4	1	2	6	70	20	10	30	20	150
5	EE-3005	ELECTRICAL CIRCUITS	4	1	2	6	70	20	10	30	20	150
6	EE-3006	PA			2							
		Total	20	5	10	30	350	100	50	150	100	750

Theory Credits: 25
Practical Credits: 05
Total Credits: 30

Theory Marks: 350
Practical Marks: 150
Quiz, Mid Term, Lab. Work: 250
Total: 750



BHABHA UNIVERSITY, BHOPAL

**COURSE CURRICULUM
FACULTY OF ENGINEERING**

**DIPLOMA IN ELECTRICAL
ENGINEERING**

(Courses effective from Academic Year 2019-20)

**SYLLABUS OF COURSES TO BE OFFERED
“DIPLOMA IN ENGINEERING”**

Disclaimer: The postgraduate programme in Diploma in Engineering syllabus & schemes is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

Dean faculty of Engineering, Bhabha University

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**DIPLOMA IN ELECTRICAL
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Branch - Electrical Engineering**

EE-3001 ELECTRICAL MACHINES-I

UNIT-I

Energy Conversion Principle - Law of conservation of energy, electromechanical energy conversion classification of machines

UNIT-II

D. C. Generator - Principle, construction, armature winding, types of winding, EMF equation, armature reaction and commutation, interpoles and compensating winding. Types of generators, characteristics and applications, losses and efficiency. Simple numerical.

UNIT-III

Single phase transformers - Principle, construction, classification. EMF equation, turns ratio, name plate rating, phasor diagram, no load and on load equivalent circuit. Voltage regulation, polarity ratio, open and short circuit tests, losses and efficiency, condition of maximum efficiency. All day efficiency and its numerical. Auto transformer. Parallel operation of single phase transformer Linear Induction Motors, construction, performance, thrust-speed characteristic, application, end effect.

UNIT-IV

Three phase transformer - Connections, groups, Scott and open delta connection. Comparison of three phase transformer with bank of three single phase transformers. Parallel operation.

UNIT-V

D. C. Motors -Principle, production of back EMF, torque equation. Classification, characteristics of D. C. motors, starters, speed control, losses and efficiency, applications of motors. Brake test, Swinburn test. Simple numericals.

**DIPLOMA IN ELECTRICAL
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LIST OF EXPERIMENTS:-

1. Study of D. C. Machines (Parts)
- 2 .Speed control of D. C. Motor (armature and field control method)
- 3 .To perform Swinburne test of DC Motor.
- 4 .Study of transformer (Parts) (single and three phases)
- 5 .To perform polarity test of single phase transformer.
- 6 .To perform ratio test of single phase transformer.
- 7 .To perform open circuit test of single phase transformer.
- 8 .To perform short circuit test of single phase transformer.
- 9 .Parallel operation of single phase transformer.

References Books:-

1. Electrical Technology Vol. II by B. L. Thareja Khanna Publisher
2. Electrical Machines by Bhattacharya, T.T.T.I.
3. Electrical Machines by Nagrath & Kothari, PHI Publication
4. Electrical Machines Vol. I & II by P.S. Bhimbra, Khanna publishers
5. fo|qr e'khusa ,e-ds-fM;ksfM;k fgUnh xzaFk vdkneh
6. oS|qr e'khusa ,p-,l-jk; nhid izdk'ku

**DIPLOMA IN ELECTRICAL
ENGINEERING**

Branch - Electrical Engineering

**EE-3002 ELECT. & ELECTRONICS MEASUREMENT AND MEASURING
INSTRUMENTS**

UNIT-I

Classification of measuring instruments, indicating, recording and integrating types of meters. Errors and types of errors, accuracy, Precision and sensitivity, Electrical measuring instruments - Construction, operation. Deflecting, controlling and damping forces, supporting systems, moving coil, electro dynamometer, moving iron and induction type instruments, simple numerical. Hot wire type instruments, vibration galvanometer, shunt and multipliers, CT & PT

UNIT-II

Wattmeter and Energy meters – Dynamometer and induction type wattmeter, Induction type energy meters. measurement of 1-phase and 3-phase power in balanced and unbalanced load condition, 3 phase wattmeter. Measurement of resistance – Classification of resistance, measurement of low, medium, and high resistance. Kelvin's double bridge, wheat-stone bridge, Ammeter, voltmeter method and ohmmeter, multimeter, megger. Importance of earth resistance, Earth tester.

UNIT-III

A. C. Bridges – Measurement of inductance and capacitance by A.C. bridges. Maxwell, Anderson, Hays, Desauty and Wien's bridge (no phasor diagram) Additional measuring instruments – Electrical resonance, Weston and vibration reed frequency meter, dynamometer power factor meter, Weston synchroscope, Merz price maximum demand meter, Rotating type phase sequence indicator

UNIT-IV

Magnetic measurement – Balastic galvanometer, measurement of flux by B.G. Gressort fluxmeter, determination of hysteresis loop for ring and bar specimen. Dielectric measurement. Meaning of dielectric loss, its importance, methods of measurement of dielectric loss by Wattmeter, C.R.O. Schering bridge.

UNIT-V

Cathode Ray Oscilloscope – CRT, Electrostatic and magnetic deflection, time base X and Y amplifiers, controls on the C.R.O. Dual beam oscilloscope. Digital storage and multi-channel CRO. Electronic Instruments - Transistor volt meter, FETVM, balanced bridge, specification of electronic voltmeter. Single and three phase electronic energy meters, mili-voltmeter and micro-volt meters. Digital instruments – Digital voltmeters- types, specifications. Digital multimeters. Counter / timers. Universal indicators for voltage, current, frequency, power, power factor, temperature, humidity etc. Digital tachometers (Contact & non Contact type). Digital controllers.

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LIST OF EXPERIMENTS:-

1. Measurement of low resistance by Kelvinn Double bridge.
- 2 .Measurement of medium resistance by wheat stone bridge.
- 3 .Measurement of insulation resistance by Megger.
- 4 .Measurement of inductance by Maxwell's bridge.
- 5 .Calibration of Voltmeter, Ammeter, Wattmeter, Energy meter.
- 6 .Measurement of P.F. by ammeter, voltmeter and wattmeter method.
- 7 .Plot B.H. curve by method of reversal using B.G.
- 8 .Use of CRO for measurement of voltage, current, phase and frequency etc.
- 9 .Measurement of 3-phase power by two wattmeter method.
10. Study and use of digital instruments, e.g. digital multi meter, frequency meter, electronic timers and counters.
- 11 .Study and use of various electrical instruments e.g. phase sequence meter, wave meter. M.D. meter, tong tester.
- 12 .Study and use of C.T. & P.T. for extension of instrument range.
- 13 .Use of multi meter in a circuit for measurement of voltage, current and resistance.

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**DIPLOMA IN ELECTRICAL
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Branch - Electrical Engineering

EE- 3003 BASIC ELECTRICAL ENGG. & MATERIALS

UNIT-I

A.C. Fundamentals - Concept of inductance, capacitance, reactance, impedance, admittance, phasor diagram of pure resistive, inductive and capacitive circuit. Difference between AC and DC quantities, sinusoidal waveform, frequency, time period. Instantaneous, maximum, average and RMS value, form factor.

UNIT-II

D.C. Circuits - Concept of charge, current, voltage, EMF, resistance, resistivity. Ohm's law, KCL, KVL. Series and parallel combination of resistances, star-delta= connection, star to delta and delta to star transformation.

UNIT-II

Electrical Engineering materials - Definition of conductors, insulators and semiconductors. Intrinsic and extrinsic semi conductor materials. Properties and applications of conducting, semi-conducting and insulating materials, classification of insulating materials on the basis of temperature. B-H curve, soft and hard magnetic materials. Different magnetic materials, properties and applications._

UNIT-IV

Magnetic effect of electric current -Concept of lines of force, flux, MMF, reluctance, permeability, magnetic flux density, magnetic field intensity. Analogy of electric and magnetic circuit, units. Faraday's laws of electromagnetic induction, self and mutual induction. Lenz's laws, Fleming's left and right hand rule.

UNIT-V

Heating effect of electric current - Heat produced. Work, power and energy, units.

Chemical effect of electric current-Faradays laws of electrolysis. Primary and secondary cells

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LIST OF EXPERIMENTS:-

1. Study of different types of meters/indicators, Ammeter, voltmeter, wattmeter etc.
2. Measurement of current and voltage in single phase and three phase circuit series and parallel circuit.
3. Measurement of current, voltage and power in single phase circuit.
4. Study of different types of loads i.e. resistive, inductive and capacitive load.
5. Study of multimeter.
6. Verification of ohms law.
7. Study of different types of conducting, insulating, and magnetic materials.
8. Study of different types of primary and secondary cells and batteries.

REFERENCES:-

1. Basic Electrical Engineering by Nagrath Kathari
2. Electrical Engineering Materials by TTTI Madras.
3. Basic Electrical Engineering by Jain & Jain
4. Basic Electrical Engineering by V.K. Mehta
5. izkjafHkd oS|qr vfHk;kaf=dh By ,e-,Q-dqjS'kh] nhid izdk'ku
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**DIPLOMA IN ELECTRICAL
ENGINEERING
Branch - Electrical Engineering
EE-3004 BASIC ELECTRONICS**

UNIT-I

Semiconductor Devices – (a) Concept of electronic emission – Different methods of electronic emission and their applications. (b) Diodes - Formation of PN junction, forward biasing and reverse biasing of PN junction, construction, characteristics and application of different types of diodes, Zener diode . (c) Transistor - PNP/ NPN Junction Transistors, different configurations: CB, CE, CC. Transistors Characteristics, and applications. (d) Special Semiconductor devices – Construction, symbol and application of Tunnel diode, photo diode, varactor, FET, MOSFET, UJT.

UNIT-II

Rectifiers –Single phase, half wave, full wave and bridge types of rectifiers. calculation of output voltage, average and RMS values, ripple factor and rectification efficiency. Filter, and types of filters. Regulated Power Supply - Difference between linear and switch mode power supply, regulated power supply and its limitations, series and shunt power supply using transistors, SMPS (Block diagram only), IC regulated power supply (78XX and 79XX series).

UNIT-III

Amplifiers - Principal of amplification, types of transistor amplifiers, biasing techniques, RC coupled, transformer coupled, and direct coupled amplifiers, push pull Amplifier, advantages and disadvantages, detailed study of circuit diagram, working principal and applications of above amplifiers, use of operational amplifier as comparator, multiplier, summer, integrator and differentiator Oscillators Principal of oscillation, Types of oscillators such as Hartley, Colpitts, tuned oscillator, Weign bridge oscillator: circuit diagram, principal, working & applications

UNIT-IV

Non-sinusoidal Generator - Astable, monostable and bitable multivariate circuits, principal of working and output waveforms Modulation and Demodulation Amplitude, phase and frequency modulation, principle, methods and applications of above modulations, A.M. and F.M. detection

UNIT-V

Integrated circuits - Concepts of IC's classification, types and their advantages, applications of common IC's such as 741, 555, 810 and digital IC's. Digital Techniques - Number system, binary, decimal number system. Addition, subtraction, multiplication & division of binary numbers. Logic gates- their symbols, truth table and applications.

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LIST OF EXPERIMENTS:-

1. Study of C.R.O. & multimeter
2. Study of electrical and electronic components
3. Colour coding of Resistors
4. Testing of Diode and Transistor
5. Study of half wave rectifier, full wave rectifier, bridge rectifier with and without filter
6. Study of Zener regulated power supply
7. Study of IC Regulated power supply (78XX and 79XX)
8. Study of transistor characteristics
9. To plot the characteristics of diode
10. To plot the characteristics of Zener diode
11. Study of transistor amplifier
12. Study of oscillator
13. Study of astable and monostable multivibrators using transistors and IC 555
14. Study of inverting and non-inverting amplifiers using IC 741 and calculation of its gain
15. Study of A.M. modulation:

REFERENCES:-

1. Basic Electronics & Linear Circuits - : By Bhargava , T.T.T.I. Chandigarh .
2. Basic Electronics -: By V.K. Mehta
3. Electronics Principles - : By Mahata.
4. Digital Electronics -: By Mahino & Leach .
5. Electronics Devices & Circuits -: By G.K. Mithal

**DIPLOMA IN ELECTRICAL
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EE-3005 ELECTRICAL CIRCUITS**

UNIT-I

CIRCUIT ANALYSIS -Active and passive elements, ideal current source and voltage source. Unilateral and bilateral elements. Number of loops, nodes, branches of a network. Analysis of networks by "Mesh" and "Node" methods. T and Π terminal networks, input and output impedance and admittance

UNIT-II

SINGLE PHASE A.C. CIRCUITS-Representation of A.C. quantity by phasor methods, rectangular and polar co-ordinates. RLC series and parallel combinations. Impedance, power in single phase circuits. Concept of power factor, conductance, admittance and susceptance. Series and parallel circuits, resonance in series circuit.

UNIT-III

NETWORK THEOREMS -Maxwell's loop theorem, Nodal analysis, Superposition, Thevenin's, Norton's and maximum power theorems with numerical problems

UNIT-IV

TRANSIENTS -Concept of transient, variation of current when connected to D.C. or A.C. series circuit (R.L. combination and R.C. combination). Time constant.

UNIT-V

POLYPHASE CIRCUITS-Concept of poly phase A.C. circuits, advantages over single phase. Generation of three phase voltage system. Three phase circuits, phase sequence, vector and wave diagrams. Star and delta connections, phase and line values of current and voltage, power in three phase circuits. Balanced and unbalanced systems

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LIST OF EXPERIMENTS:-

1. Verification of Superposition theorem
2. Verification of Norton's and Thevenin's theorem
3. Verification of Maximum power transfer theorem
4. Performance of R-L-C- series circuit
5. Performance of R-L-C- parallel circuit
6. Study of electrical resonance in series circuit
7. Verification of relation between line and phase voltage and current in 3-phase circuit
8. Study of transients